Namal University, Mianwali

Software Requirements Specification (SRS)

Project: Point-of-Sale System

Course: CSC-225 – Software Engineering

Department: Computer Science

Semester: Fall 2025

Submitted To:
Ms. Asiya Batool
Lecturer, Department of Computer Science

Submitted By:

Muhammad Shoaib	Reg No: Num-Bscs-2024-57
Allah Ditta	Reg No: Num-Bscs-2024-10
Zainab Bibi	Reg No: Num-Bscs-2024-79

Submission Date: 28 Dec 2025

Contents

1 introduction

1.1 Purpose

The purpose of this SRS is to formally define the requirements of the Point-of-Sale (POS) system for a pizza shop.
This document serves as a reference for developers, administrators, and end users to understand the system's intended functionality and constraints.
The SRS follows IEEE 830-1984 to ensure clarity, completeness, and consistency.

1.2 Scope

The POS system is the software product to be developed.
It shall support restaurant operations including order placement, deal recommendations, billing, kitchen coordination, and administrative management.
The system shall support multiple user roles: Waiter, Cashier, Kitchen Staff, and Admin.
The goals of the system include:

- Improving order accuracy
- Reducing service time
- Providing real-time visibility of order status across all operational roles

1.3 Definitions, Acronyms, and Abbreviations

Term	**Description**
POS	Point-of-Sale system used for sales and order processing
SRS	Software Requirements Specification
RP	Requirement Provider
Scrum	Agile development framework used for iterative development and team collaboration
Sprint	Time-boxed iteration in Scrum methodology

Table 1: Definitions, Acronyms, and Abbreviations

1.4 References

1. IEEE Std 830-1984: Software Requirements Specification, IEEE, 1984.
2. Team meeting with Requirement Provider (RP) regarding POS system requirements, December 2025.

1.5 Overview

This document provides a high-level description of the POS system.
General system characteristics, user roles, constraints, and assumptions are described in the following sections.
Specific functional and non-functional requirements, interface requirements, and system attributes are detailed in subsequent sections.

2 General Description

2.1 Product Perspective

The POS system is a centralized software application deployed on devices within the restaurant.
It replaces manual order handling and disconnected billing systems.
The system integrates order processing, kitchen notifications, and billing into a single platform with role-based user interfaces.
Supported hardware includes desktops, and tablets.

2.2 Product Functions

The system provides the following functions at a high level:

- Role-based login for Waiter, Cashier, Kitchen Staff, and Admin
- Menu display and order placement
- Automatic deal recommendation for cost optimization
- Real-time order status tracking
- Bill generation and payment handling
- Administrative reporting and menu management

2.3 User Characteristics

- **Waiter:** Places orders, selects tables, views order status, serves customers.
- **Cashier:** Generates bills, handles payments, views order summaries.
- **Kitchen Staff:** Receives and updates order preparation status.
- **Admin:** Manages menu items, views sales reports, and controls system data.

Users are assumed to have basic knowledge of POS systems.

2.4 General Constraints

- The system shall follow Scrum methodology for development.
- The SRS must comply with IEEE 830-1984 standards.
- Hardware and network limitations, interface compatibility, and security considerations shall be addressed in the Specific Requirements section.

2.5 Assumptions and Dependencies

- Users have basic POS knowledge.
- Required hardware devices (Desktops, tablets, printers, cash drawers) are available and functional.
- Payment gateways are provided by third-party services.
- Kitchen and cashier terminals are connected to the same network.

3 Specific Requirements

3.1 Functional Requirements

3.1.1 User Authentication

Functional Requirements:

- **FR-01.1** The system shall allow authorized users to log in using a unique username and password.
- **FR-01.2** The system shall validate the entered credentials against stored user records.
- **FR-01.3** The system shall grant access based on the authenticated user's role.
- **FR-01.4** The system shall deny access and display an error message if authentication fails.

Introduction: System ensures that only authorized users can access the system and that users interact only with features relevant to their role.

Inputs:

- **Data:** Username and password.
- **Sources:** Login interface on devices (tablet, desktop).
- **Quantities:** One username and password pair per login.
- **Units of Measure:** Strings (username and password).

- **Timing:** On Login attempt.
- **Valid Range:** Username and password strings of 8–16 alphabets, numeric and a special characters.
- **Operator Controls:** User add username and password and clicks “Login”.
- **Interface References:** GUI login screen design.

Processing:

- Verify input details against stored data.
- Check role and assign permissions after successful authentication.
- Deny access and display error message if invalid credentials.
- **Abnormal Situations:** Invalid login, communication failure with database.
- **Parameters Affected:** User login status, role permissions.
- **Methods:** Check password securely.
- **Degraded Operation:** If database unavailable, allow offline login for limited access.

Outputs:

- **Data:** Access granted or show a error message.
- **Timing:** Immediate after login attempt.
- **Valid Range:** Success or failure states.
- **Illegal Values:** Display error message for invalid login details.
- **Interface References:** GUI for successful and unsuccessful login.

3.1.2 Order Placement

Functional Requirements:

- **FR-02.1** The system shall allow the waiter to select a table number for a new order.
- **FR-02.2** The system shall display the available menu items to the waiter after table selection.
- **FR-02.3** The system shall allow the waiter to add one or more menu items to an order.
- **FR-02.4** The system shall allow the waiter to update or remove selected menu items before order confirmation.
- **FR-02.6** The system shall generate and display an order summary upon confirmation.

- **FR-02.7** The system shall forward the confirmed order to the cashier and kitchen staff.

Introduction: Software allow the waiter to take customer orders accurately, select table numbers, add menu items.

Inputs:

- **Data:** Table number, menu items.
- **Sources:** Waiter device interface.
- **Quantities:** One table number, multiple items per order.
- **Units of Measure:** Integer (table number, quantity), currency (item price).
- **Timing:** During customer ordering.
- **Valid Range:** Table number 1–50; quantity>0; item prices>0.
- **Operator Controls:** Touchscreen to select table, items, quantities, and confirm order.
- **Interface References:** GUI menu screen, deal suggestion popup.

Processing:

- Display menu after table selection.
- Add, update, or remove items in order.
- Automatically check for compatible deals.
- Validate item selection and quantity.
- Confirm and submit order.
- **Abnormal Situations:** Item selected by customer is unavailable, quantity exceeds stock, communication failure.
- **Parameters Affected:** Order contents, total price, status.
- **Methods:** Check order and apply deal.

Outputs:

- **Data:** Order summary with table number, items or deal, quantities.
- **Destinations:** Waiter, cashier and kitchen staff devices.
- **Timing:** Immediately after confirmation.
- **Valid Range:** Order totals > 0.
- **Illegal Values:** Display error if item or quantity invalid.
- **Interface References:** GUI order summary screen.

3.1.3 Order Processing

Functional Requirements:

- **FR-03.1** The system shall assign a unique order ID to each confirmed order.
- **FR-03.2** The system shall transmit the confirmed order details to the cashier and kitchen staff.
- **FR-03.3** The system shall set the initial status of a new order to *Order Pending*.
- **FR-03.4** The system shall track and update the order status throughout its lifecycle.
- **FR-03.5** The system shall display updated order status information to the waiter, cashier, and kitchen staff.

Introduction: The Software sends orders to the cashier and kitchen and keeps track of their status.

Inputs:

- **Data:** Confirmed order (items or deal, quantities, table number).
- **Sources:** Waiter device.
- **Quantities:** One order per table.
- **Units of Measure:** string (Items), integer (quantity), currency (price).
- **Timing:** Immediately after order submission.
- **Valid Range:** order list can not be empty.
- **Operator Controls:** None because system automatic process it.
- **Interface References:** Popups for kitchen staff and order display for cashier.

Processing:

- Assign unique order ID.
- Send order summary to cashier and kitchen staff.
- Set initial order status to Order Pending.
- Update status in real time.
- **Abnormal Situations:** Network failure, duplicate order ID.
- **Parameters Affected:** Order status, order ID.
- **Methods:** Update order status automatically and assign unique IDs.

Outputs:

- **Data:** Order summary, status updates.
- **Destinations:** Cashier and kitchen staff devices, waiter device.
- **Timing:** Real time.
- **Valid Range:** Status: Order Pending, In Preparation, Prepared, Served, Completed.
- **Illegal Values:** Error notifications if order has an invalid status.
- **Interface References:** Dashboard and notification GUI for cashier, kitchen staff, and waiter.

3.1.4 Kitchen Operations

Functional Requirements:

- **FR-04.1** The system shall display all pending orders to the kitchen staff.
- **FR-04.2** The system shall allow kitchen staff to accept an assigned order for preparation.
- **FR-04.3** The system shall update the order status when preparation begins.
- **FR-04.4** The system shall allow kitchen staff to mark an order as prepared upon completion.
- **FR-04.5** The system shall notify the corresponding waiter when an order is marked as prepared.

Introduction: The software Supports kitchen staff in preparing orders and notifying the waiter on completion.

Inputs:

- **Data:** Pending orders, acceptance, completion updates.
- **Sources:** Kitchen staff device.
- **Quantities:** One order or multiple orders displayed to kitchen staff.
- **Units of Measure:** strings (Status),integers (order ID).
- **Timing:** During order preparation.
- **Valid Range:** Status values: Pending → In Preparation → Prepared.
- **Operator Controls:** Touchscreen or Desktop to accept or complete orders.
- **Interface References:** Kitchen dashboard GUI.

Processing:

- Display pending orders.
- Update order status on acceptance and completion.
- Notify corresponding waiter when order is completed.
- **Abnormal Situations:** Order canceled, system crash, missing order ID.
- **Parameters Affected:** Order status, notifications.
- **Methods:** Update order status automatically.

Outputs:

- **Data:** Updated order status, waiter notified.
- **Destinations:** Kitchen dashboard, waiter interface.
- **Timing:** Real-time.
- **Valid Range:** Status: Pending → In Preparation → Prepared.
- **Illegal Values:** Error displayed on dashboard if status is invalid .
- **Interface References:** Dashboard and Notification popup GUI.

3.1.5 Order Serving

Functional Requirements:

- **FR-05.1** The system shall notify the waiter when an order is marked as prepared by the kitchen staff.
- **FR-05.2** The system shall allow the waiter to mark an order as served to the customer.
- **FR-05.3** The system shall update the order status to Served when the waiter confirms serving.
- **FR-05.4** The system shall allow the waiter to mark an order as completed after the customer finishes the meal.
- **FR-05.5** The system shall update the order status to Completed and make it visible to cashier and admin dashboards.

Introduction: The software tracks the serving of orders and updates the status after the order is served to the customer and consumed.

Inputs:

- **Data:** Order prepared notification from kitchen.
- **Sources:** Kitchen staff device (Prepared) and waiter device.

- **Quantities:** One order per table.
- **Units of Measure:** Strings (Status).
- **Timing:** After preparation and during serving.
- **Valid Range:** Status values: Served $\rightarrow$ Completed.
- **Operator Controls:** Touchscreen or keyboard to mark order as served or completed.
- **Interface References:** Waiter order screen GUI.

Processing:

- Notify waiter when order is prepared and ready.
- Waiter serves the order and updates status to Served.
- Update status to Completed after customer finishes.
- **Abnormal Situations:** Missed notification, incorrect table selection.

Outputs:

- **Data:** Updated order status.
- **Destinations:** Waiter, cashier, and admin dashboards.
- **Timing:** Real-time.
- **Valid Range:** Status: Served $\rightarrow$ Completed.
- **Illegal Values:** Display error to waiter.
- **Interface References:** Status update GUI.

3.1.6 Billing and Payment

Functional Requirements:

- **FR-06.1** The system shall generate a bill for each completed order including item prices, taxes, service charges, and discounts.
- **FR-06.2** The system shall allow the cashier to select a payment method (cash or online payment).
- **FR-06.3** The system shall validate the payment amount against the total bill amount.
- **FR-06.4** The system shall process payments securely and update the payment status.
- **FR-06.5** The system shall generate and store a payment receipt for accounting and audit purposes.

Introduction: The software handles bill generation and payment processing, supporting both cash and online payments, ensuring accurate price totals and secure transactions.

Inputs:

- **Data:** Order summary (items, quantities, prices), selected payment method, payment amount provided by customer.
- **Sources:** Cashier device or online payment interface.
- **Quantities:** One payment per completed order.
- **Units of Measure:** Currency (Bill).
- **Timing:** After order completion.
- **Valid Range:** Payment amount $\geq$ order total including taxes and discounts.
- **Operator Controls:** Cash entry, card swipe, or online payment confirmation via GUI.
- **Interface References:** Billing and payment screen GUI.

Processing:

- Calculate the total bill including taxes, service charges, and applied discounts.
- Validate the payment amount against the order total.
- Process payment through the selected method and update payment status.
- Store transaction records securely for accounting and auditing.
- **Abnormal Situations:** Insufficient payment, network failure during online payment, or transaction declined.
- **Parameters Affected:** Payment status, order records, transaction logs.
- **Methods:** Arithmetic calculations for totals, payment validation, secure transaction.

Outputs:

- **Data:** Bill, payment confirmation receipt.
- **Destinations:** Cashier interface, admin dashboard for records.
- **Timing:** Immediately after successful payment.
- **Valid Range:** Total amount > 0.
- **Illegal Values:** Error messages for invalid payment or system failures.
- **Interface References:** Billing, payment, and receipt GUI screens.

3.1.7 Administration

Functional Requirements:

- **FR-07.1** The system shall generate detailed sales reports (daily, weekly, monthly) upon admin request or schedule.
- **FR-07.2** The system shall identify and display most-selling and low-performing menu items for analysis.
- **FR-07.3** The system shall allow the admin to add, update, or remove menu items.
- **FR-07.4** The system shall allow the admin to manage user accounts, including adding, updating, deactivating, or removing users.
- **FR-07.5** The system shall display error messages for invalid input, duplicate entries, or failed operations.

Introduction: The Software allows admin to monitor restaurant performance, managing menu items, and controlling user accounts.

Inputs:

- **Data:** Requests for sales reports, menu updates, and user management.
- **Sources:** Admin interface.
- **Quantities:** One action per request (report generation, menu modification, or user account management).
- **Units of Measure:** string (menu item , reports).
- **Timing:** On demand or scheduled intervals.
- **Valid Range:** sales ≥ 0.
- **Operator Controls:** Admin clicks.
- **Interface References:** Admin dashboard GUI.

Processing:

- Generate detailed sales reports (daily, weekly, monthly).
- Identify most-selling items and low-performing items for analysis.
- Add, update, or remove menu items.
- Manage user accounts: add, update, deactivate, or remove users.
- **Abnormal Situations:** Invalid input data, system crash, duplicate menu entries.
- **Parameters Affected:** Menu database, sales report data, user account records.

Outputs:

- **Data:** Generated reports, updated menu items, user account changes.
- **Destinations:** Admin dashboard, waiter menu interface.
- **Timing:** Real time updates for changes, scheduled outputs for reports.
- **Valid Range:** Accurate sales detail, valid menu item details, correct user account detail.
- **Illegal Values:** Error messages displayed for invalid input, failed operations, or duplicates.
- **Interface References:** Admin dashboard GUI.

3.2 Performance Requirements

This subsection specifies both static and dynamic numerical requirements placed on the software or on human interaction with the software.

3.2.1 Static Numerical Requirements

- The system shall support at least 50 concurrent users (waiters, cashiers, kitchen staff, admins).
- The system shall handle 50 simultaneous users without performance degradation.
- The system shall support processing up to 500 orders per day.
- The system shall maintain 99.99% consistency of all order, payment, and report records for up to 500 daily transactions, verified by automated integrity checks at least once per hour.

3.2.2 Dynamic Numerical Requirements

- The system shall load menu within 3 seconds.
- The system shall process orders entered by the waiter and update the status to the cashier and kitchen staff within 2 seconds per order.
- The system shall reflect order status updates in all relevant user interfaces in real time with a maximum latency of 1 second.
- The system shall generate reports for daily sales within 5 seconds and for up to one year of data within 10 seconds.
- The system shall complete and log payment transactions successfully at least 99.99% of the time.

- The system shall allow new users to learn and perform primary tasks (login, place/update orders, generate reports) within 10 minutes of training, for at least 90% of users tested.
- The system shall provide visual notifications (pop-ups) for order updates within 1 second of the triggering event.

3.3 Design Constraints

3.3.1 Standards Compliance

- The system shall follow standard rules for report formats, data naming, and review records.
- The system shall record all changes in orders, payments, or reports with before and after values, and these records shall be kept for at least 12 months.
- The system shall produce reports within 5 seconds for daily report and within 10 seconds for upto 1 year when requested.

3.3.2 Hardware Limitations

- The system shall run on Windows 10 or newer and Android 10 or newer.
- The system shall work on PCs with at least 4 GB RAM, 2-core CPU, and 256 GB storage, and on Android devices with at least 2 GB RAM and 16 GB storage.
- The system shall process all orders, payments, and reports without using more than 80% of CPU or memory on peak load.
- The system shall work properly with network speeds of at least 10 Mbps for proper operation across multiple devices.

3.4 Attributes

3.4.1 Availability

- The system shall be 99.99% available during restaurant hours for example (10:00 AM–11:00 PM).
- The system shall have a maximum downtime of 2 hours per month for maintenance.
- The system shall maintain 99.5% uptime for critical services (login, order processing, payment) during operating hours.

3.4.2 Security

- Each user shall access the system only after successful authentication.
- Admin operations shall require multi factor authentication.
- The system shall log and report unauthorized access attempts within 10 seconds.

3.4.3 Maintainability

- The system shall reflect updates to menu items or deals within 5 seconds of admin confirmation.
- The system shall allow incremental updates for new features without downtime exceeding 10 minutes.

3.4.4 Portability / Transferability

- The system shall operate on Windows 10 or newer and Android 10 or newer.
- The system shall maintain consistent user interface and functionality across supported platforms.

3.4.5 Data Integrity

- The system shall maintain 100% consistency of order, payment, and report data for up to 500 daily transactions.
- In case of power failure or system crash, the system shall restore the last consistent state within 30 seconds of reboot.

3.5 External Interface Requirements

3.5.1 User Interfaces

- The system shall provide separate graphical user interfaces for Waiter, Cashier, Kitchen Staff, and Admin.
- Interfaces shall allow login, order placement, order tracking, bill generation, and reporting according to user roles.
- The GUI shall include touch friendly elements for android use.
- Interfaces shall provide visual notifications (pop-ups) for order updates, errors, and confirmations within 1 second of the event.
- Each interface shall use consistent screen formats and layouts.
- New users shall be able to perform primary tasks (login, place/update orders, generate reports) within 10 minutes of training, verified with at least 90% success rate in test users.

3.5.2 Hardware Interfaces (6.3.1.5.2)

- The system shall run on desktop PCs, and Android.
- The system shall interface with receipt printers for printing bills and order receipts.
- Cash drawers shall open automatically after successful cash payment.
- The system shall support standard screen resolutions of at least 1280×720 for PCs and tablets.

3.5.3 Software Interfaces

- The system shall integrate with the restaurant database for storing user accounts, orders, payments, and menu items.
- The system shall integrate with payment gateways (online payment APIs) to process digital payments securely.

3.5.4 Communications Interfaces

- The system shall communicate over the local area network (LAN) to synchronize orders across devices in real time.
- Internet connectivity shall be used for online payment processing and admin access.

3.6 Other Requirements

3.6.1 Database Requirements

- The database shall store orders, payments, user accounts, and menu items.
- Data shall be accessed quickly so that order status and reports are available in real time.
- The system shall let at least 50 users access and update data at the same time without errors or delays.
- Data elements and records shall follow defined field types, lengths, and relationships.
- The database shall be organized to handle up to 500 transactions per day without errors or performance problems.
- Dynamic organization: system shall maintain real-time updates for orders, payments, and reports.
- Data shall be retained for at least 12 months until admin specified.
- Cash drawers shall open automatically after successful cash payment.

A Use Case Diagram

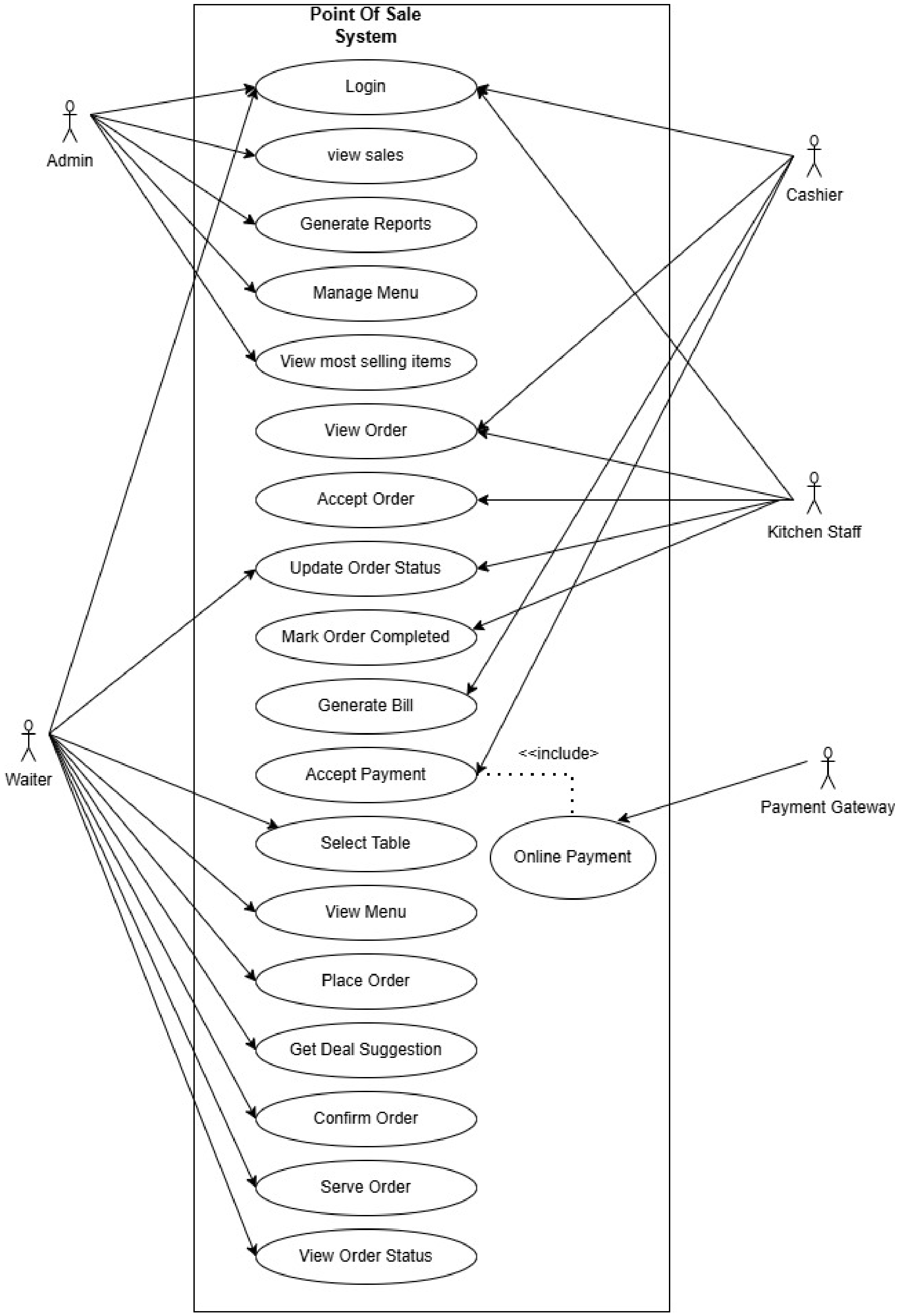

Figure 1: Use Case Diagram of the POS System

B Context Diagram

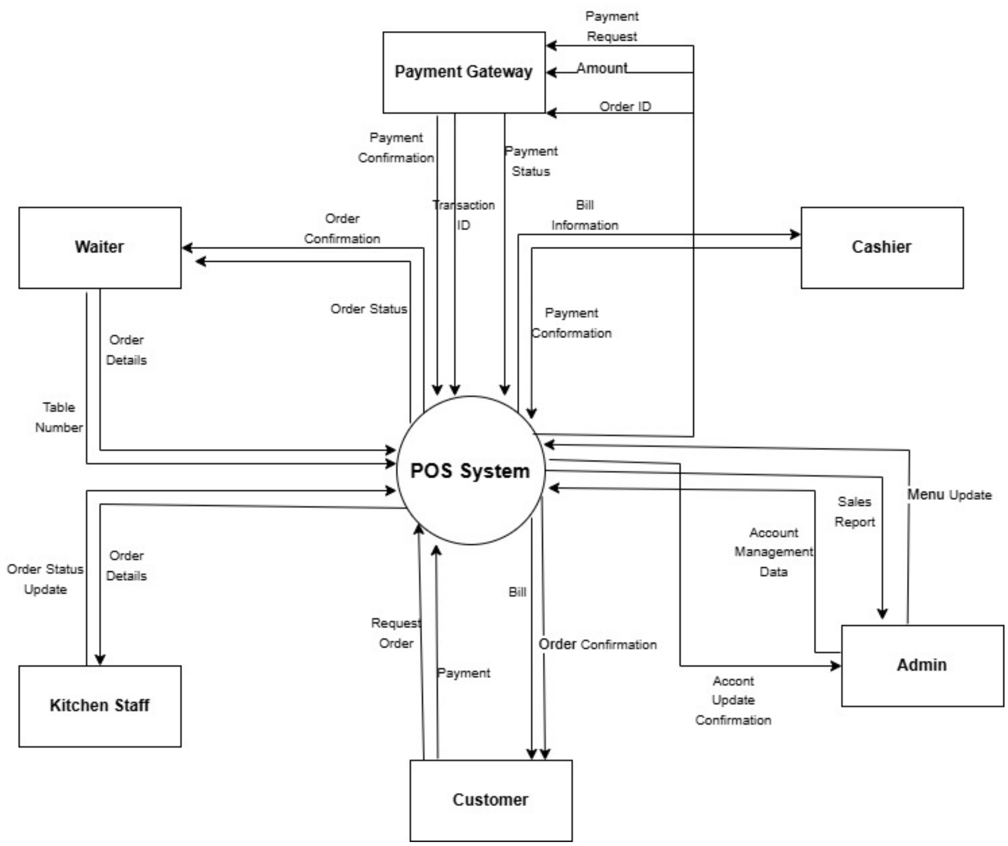

Figure 2: Context Diagram of the POS System